

SH Crocs Research Proposal: Optimizing experimental conditions for recombinant synthesis of Bacterial Cellulose in *E. coli* through the overexpression of phosphoglucomutase and UGPase

Abstract

Bacterial cellulose has been gaining prominence due to its limited environmental burden and its superior chemical properties. Although native in *Komagataeibacter* strains, with *K. xylinus* and *hanseii* identified as natural strong producers, this process is rarely scalable due to low productivity, slow bacterial growth, and the spontaneous mutation of cellulose-producing bacteria into cellulose non-producing bacteria. With these concerns, recombinant BC synthesis in *E. coli* has been studied and has shown promising yield. As a potential way to increase yield while minimizing metabolic burden on cell strains, this project investigates overexpressing native phosphoglucomutase and UGPase. Three promoters, *t7lac*, *pCspA*, and *araBad*, are tested in varying temperatures and inducer concentrations, and the most suitable combination given BC production will be identified. After this, a theoretical pathway that combines the BC synthase operon, accessory genes *CmcpX* and *CcpX*, and the most suitable promoter combination will be designed.

1. Literature Review

1.1 Overview and Significance of Bacterial Cellulose

Cellulose, a structural polysaccharide found in plant cell walls and produced by certain bacteria, is one of the most common organic materials on Earth (Naomi et al., 2020). Among cellulose variants, bacterial cellulose (BC) is gaining prominence due to its excellent physicochemical and biological properties. Although its chemical backbone is identical to that of plant cellulose, BC displays markedly superior physical properties because it is produced as a highly pure polymer (free of lignin and hemicellulose) and exhibits high crystallinity (up to 90%) (Ul-Islam et al., 2019; Popa et al., 2022). Its structure consists of bundles of cellulose nanofibrils (2-4 nm), which assemble into ribbon-like fibers approximately 100 µm in length and diameter, fibers much smaller than plant cellulose. This distinctive nanofibrillar arrangement makes BC much more porous, ensuring a higher water-retaining capacity and hydrophilicity, conferring enhanced elasticity, flexibility, and stability under humid conditions. The combination of these structural and chemical characteristics, including high polymerization degree, hydrophilicity, moldability, and non-toxicity, makes BC a uniquely versatile biomaterial suitable for biomedical and technological applications (Popa et al., 2022).



Figure 1. *Advantages of bacterial cellulose* (Popa et al., 2022)

In biomedical applications, BC has been explored for wound dressing, bone repair, controlled drug delivery, and tissue engineering due to its biocompatibility, non-toxicity, and ability to support cell adhesion and proliferation (Abeer et al., 2014; Tronser et al., 2018; Pang et al., 2020). Beyond biomedicine, BC has shown promise in sustainable packaging and flexible electronics, where its nanoscale structure enhances strength and conductivity (Trigona et al., 2019; Sid et al., 2021). These properties and applications make BC an essential focus in synthetic biology as researchers aim to engineer scalable, efficient production systems.

1.2 Metabolic Pathway and Role of UDP-Glucose

BC synthesis begins with the conversion of glucose into activated sugar intermediates that feed directly into cellulose synthesis. In cellulose-producing bacteria, such as *K. xylinus*, glucose is first phosphorylated to glucose-6-phosphate by glucokinase, then converted to glucose-1-phosphate (G1P) by phosphoglucomutase (Singhania et al., 2021). UDP-glucose pyrophosphorylase (UGPase) subsequently catalyzes the reaction between glucose-1-phosphate (G1P) and UTP to synthesize UDP-glucose (UDP-Glc). UDP-Glc is then polymerized into cellulose by the bacterial cellulose synthase complex encoded by the *bcs* operon and secreted extracellularly. The intracellular availability of UDP-Glc is therefore a critical factor in the cellulose biosynthetic pathway, as it serves as the direct substrate for cellulose polymerization.

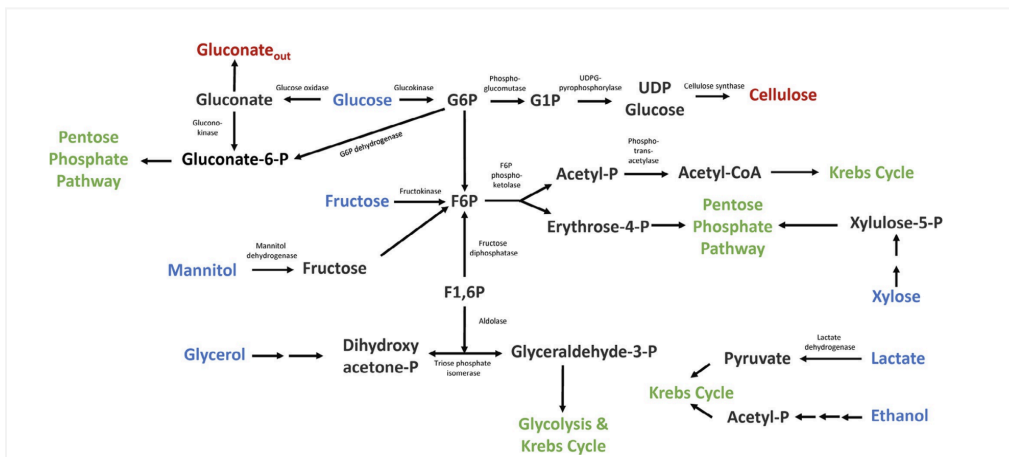


Figure 2. *Metabolic pathways for different substrates (in blue) in K. xylinus* (Raghavendran et al., 2020)

The specific mechanisms by which cellulose polymerization and export are mediated by the bacterial cellulose synthase complex, encoded by the *bcsABCD* operon (Singhania et al., 2021). This operon codes for four core proteins: BcsA, BcsB, BcsC, and BcsD. All four subunits are required for effective cellulose synthesis *in vivo*, although *in vitro* studies indicate that only BcsA and BcsB are essential (Malcı et al., 2024). BcsA is a membrane protein, allosterically activated by c-di-GMP (cyclic diguanylate monophosphate), that functions as the catalytic subunit of cellulose synthase, while BcsB serves as a regulatory subunit (Swinger et al., 2021). Although the roles of BcsC and BcsD are less well understood, cellulose is translocated to the outer membrane through a pore formed by BcsC; BcsD seems to assist in the proper orientation of the linear terminal complexes along the longitudinal axis of the cell (Acheson et al., 2022).

In addition to the core *bcs* genes, several accessory enzymes regulate cellulose synthesis and secretion. The *ccpAx* gene encodes a short polypeptide that binds to BcsD and influences crystalline cellulose secretion *in vivo* (Abidi et al., 2022). *CcmAx* encodes an endo- β -1,4-glucanase with cellulose-hydrolyzing activity that enhances cellulose synthesis. BcsZ helps prevent periplasmic clogging during cellulose export; knockout of *bcsZ* has been shown to significantly reduce BC yield in *K. xylinus* (Li et al., 2021).

1.3 Bacterial Cellulose Biosynthesis

BC is naturally produced by several gram-negative bacteria. Certain strains synthesize cellulose as an essential component of biofilm formation, with *Komagataeibacter xylinus* and *Komagataeibacter hansenii* among the most efficient native producers (Abidi et al., 2021; Buldum et al., 2021). Due to BC's high crystallinity, purity, and water-retaining capacity, there is significant demand for BC in biomedical and bioplastic applications. However, industrial-scale production remains limited by several factors, including 1) slow growth compared to other bacteria, 2) low productivity, 3) sensitivity to culture conditions, and 4) spontaneous mutation of cellulose-producing strains into non-producing variants (Buldum et al., 2018). Despite these challenges, substantial progress has been made in adapting *Komagataeibacter* strains for large-scale BC production. Recent genetic engineering approaches have established genetic toolkits in *K. xylinus*, positioning it as a model organism for bacterial cellulose synthesis and enabling more precise manipulation of BC production (Florea et al., 2016; Teh et al., 2019).

Given BC's broad applicability in biomedicine, several studies have explored engineering cellulose properties through synthetic biology approaches (Fang et al., 2015; Fillat et al., 2018; Jacek et al., 2019). For example, co-culturing *K. rhaeticus* with engineered strains of *Saccharomyces cerevisiae* enabled the production of yeast-embedded cellulose with applications in biosensing (Gilbert et al., 2021). In an *Enterobacter* strain, overexpression of colonic acid biosynthesis genes enhanced water-holding capacity by 1.7-fold (Liu et al., 2018). Additionally, Yadav et al. (2010) demonstrated that engineered *K. xylinus* strains incorporating N-acetylglucosamine residues in BC production produced a chitin-cellulose composite with increased biodegradability, highlighting potential applications in biofuel production and tissue engineering.

Genetic engineering has also been applied to enhance BC yield by minimizing byproducts or overexpressing key biosynthetic genes. As shown in Figure 2, glucose can be diverted toward gluconate and gluconic acid production, reducing carbon availability for cellulose synthesis (Raghavendran et al., 2020). Deletion of glucose dehydrogenase (*gdh*) in *K. xylinus* BPR2001 redirected carbon flux toward BC synthesis, resulting in a 1.7-fold increase in cellulose production compared to wild-type strains (Shigematsu et al., 2005). Similarly, *gdh* knockout in *K. xylinus* BCRC12334 led to a 230%

increase in BC production under shaken culture conditions. CRISPR-based editing of the accessory genes *bcsH*, *bcsX*, *bcsY*, and *bcsZ* increased BC production by 23.6% in *K. xylinus*, while elevated intracellular levels of cyclic di-GMP, an allosteric activator of BcsA, further promoted cellulose synthesis (Buldum and Mantalaris, 2021; Huang et al., 2025). Dong et al. (2020) further demonstrated that overexpression of *pgm*, *galU*, and *ndp* (nucleoside diphosphate) resulted in a 1.9-2.2-fold increase in BC yield relative to wild-type *K. xylinus*. Collectively, these studies highlight the effectiveness of genetic engineering strategies in enhancing BC production.

Due to the slow growth and genetic instability of *Komagataeibacter* strains, heterologous expression of BC biosynthesis pathways has been explored, particularly in *Escherichia coli*. The first successful report of BC synthesis in *E. coli* was described by Imai et al. (2014), in which *bcsAB* and a diguanylate cyclase were expressed in *E. coli* XL1-Blue. However, although BC production was observed, proper crystallization did not occur. Subsequent studies identified additional challenges, including toxicity associated with *bcsAB* expression, plasmid instability, and premature cell death, which limited BC production in common laboratory strains such as BL21 (Buldum et al., 2018; Al-Janabi et al., 2022). To address these issues, Buldum et al. (2018) expressed *bcsABCD* along with accessory genes *cmcAx* and *ccpAx* in *E. coli* HMS174 (DE3) and C41 (DE3), strains engineered for toxic protein expression, achieving BC yields comparable to *K. xylinus*. Al-Janabi et al. (2022) further demonstrated BC synthesis in *E. coli* DH5 α , although the authors suggested that DH10 β may offer improved productivity. They also found that accessory genes *cmcAx* and *ccpAx* were not necessary for BC synthesis. Both studies reported that recombinant *E. coli* produced cellulose fibers with larger diameters and higher crystallinity, and that BC synthesis was optimal at low temperatures and low IPTG (<0.02mM IPTG) induction levels.

Despite these advances, a clear gap in the literature remains: relatively few studies have systematically examined genetic engineering strategies to enhance bacterial cellulose synthesis in *E. coli*. Given that recombinant BC production in *E. coli* is favored at lower temperatures and that heterologously produced cellulose often exhibits higher crystallinity and mechanical strength, *E. coli* represents a promising yet underexplored chassis for industrial BC production.

1.4 Design Rationale: UGPase Optimization in *E. coli*

The requirement of UDP-Glc for cellulose synthesis has been demonstrated at the enzymatic level. In vitro studies using a purified *Rhodobacter sphaeroides* BcsA-B complex showed that cellulose biosynthetic activity strictly depends on the presence of UDP-Glc, as well as the allosteric activator cyclic di-GMP (Omadjela et al., 2013). These findings align with the current understanding of BC synthesis pathways and suggest that precursor availability plays a critical role in enabling cellulose synthase activity.

Thus, UGPase, the enzyme responsible for the catalysis of UDP-glc, plays a key role in the BC metabolic pathway. Previous studies have shown that UGPase activity is up to 100-fold higher in native BC-producing bacteria. Furthermore, the UGPase-catalyzed reaction has been proposed as a potential rate-limiting step in this pathway, indicating a metabolic bottleneck that may constrain BC production and could be targeted for optimization (Park et al., 2010; Zhang et al., 2017). In a study by Zhang et al. (2017), the team used flux balance analysis to model the effects of gene overexpression on BC biosynthesis. Although this analysis was performed in a *Komagataeibacter* strain, the study identified eight genes as potential optimization targets, with overexpression of UGPase, *pgm*, and G6PIS predicted to yield the greatest increases in BC production, as shown in Figure 3.

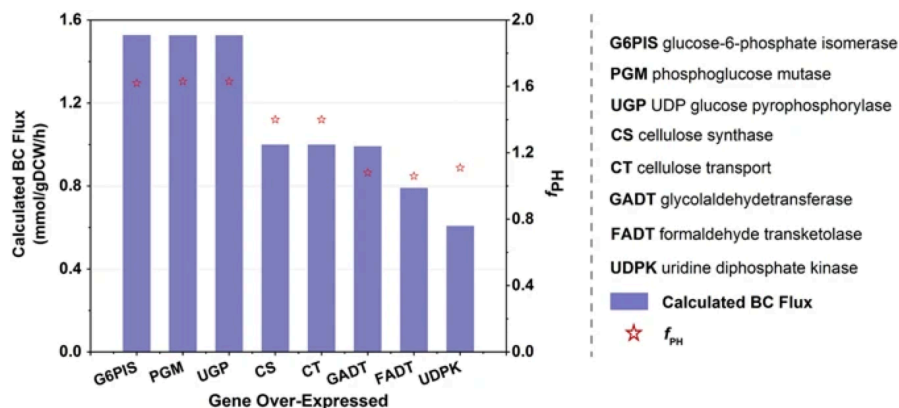


Figure 3. A graph that compares the calculated bacterial cellulose flux to f_{PH} , with different genes that were overexpressed (Zhang et al., 2017)

Our project is based on the premise that increasing intracellular UDP-Glc concentration by overexpressing UGPase could enhance BC production. Despite metabolic modeling and pathway-level analyses, there has yet to be a direct experimental study with empirical data, even in *Komagataeibacter* strains, that has concluded that increasing UDP-Glc concentrations will definitely increase BC yield. Most previous research has focused on optimizing growth conditions, gene expression, or overall metabolic flux, rather than explicitly testing the effect of UDP-Glc concentration (Pourrameza et al., 2009; Aswini et al., 2020; Zywicka et al., 2025). This gap provided the rationale for our approach in investigating the impact of UDP-Glc and UGPase in a homologous host in vivo to enhance BC production. Despite an apparent lack of literature and research about this topic, especially in expressing these genes in non-native strains of bacteria, the findings in Zhang et al. (2017) and Omadjela et al. (2013) supported our decision to move forward with this strategy, as they provided us with computational and conceptual reasoning that UDP-Glc could influence BC levels.

As noted previously, prior literature has primarily focused on recombinant gene expression in *Komagataeibacter* species, with relatively few studies examining BC-related pathway engineering in non-native hosts. Accordingly, we selected *Escherichia coli* as our chassis organism due to its well-characterized genetics, extensive molecular biology toolkit, and ease of manipulation (Tepap et al., 2023). Specifically, *E. coli* C41(DE3) was chosen for its enhanced plasmid stability and tolerance to toxic recombinant protein expression, as well as its higher transformation efficiency compared to strains such as *E. coli* BL21(DE3) (Dumon-Seignovert et al., 2004). These characteristics are particularly valuable for this project, as UGPase overexpression may impose metabolic stress or toxicity that could otherwise limit cell viability.

Using this strain and plasmid system, we aim to characterize the relationship between intracellular UDP-Glc concentration and BC production while identifying conditions that maximize UDP-Glc availability. In the first experimental phase, different promoters (T7lac, proU, AraBAD, and pCspA) will be evaluated to determine which enables the highest UGPase expression without causing excessive cellular toxicity. Once an optimal promoter is identified, culture conditions will be optimized for BC production in *E. coli* by varying carbon feedstock, incubation temperature, inducer concentration, and growth conditions (shaking versus static). Due to time constraints, a theoretical pathway for expressing the *bcsABCD* genes (encoding the core BC synthase complex) in *E. coli* will be proposed rather than experimentally implemented. This framework will nonetheless provide a basis for interpreting how UDP-Glc availability may influence BC production and for guiding future empirical studies.

However, we recognize several limitations in the design of this experiment. Most notably, it remains uncertain whether increasing intracellular UDP-Glc levels through UGPase overexpression will directly lead to increased BC production in vivo, despite existing biochemical and computational evidence suggesting this relationship. Additionally, high-level overexpression of metabolic enzymes can impose substantial physiological stress on host cells, potentially reducing growth rates or diverting resources away from BC synthesis, a challenge commonly reported in recombinant protein expression studies. Although *E. coli* C41(DE3) was deliberately selected for its tolerance to toxic protein expression and a low- to medium-copy-number plasmid backbone was used to mitigate metabolic burden, overexpression of UGPase may nonetheless result in unintended toxicity or metabolic imbalance that limits BC production.

Key Terminology/Abbreviations

Kombucha/k. - komagataeibacter/acetobacteria

UGPase: UDP-glucose pyrophosphorylase (UDPG), UTP-glucose-1-phosphate uridylyltransferase,

UDPG-pyrophosphorylase

pgm: phosphoglucomutase

galU: UTP--glucose-1-phosphate uridylyltransferase

UTP: Uridine Triphosphate

2. Genetic Construct, Source Documentation, and Plasmid Design

2.1 Materials

The following materials are used: *Escherichia coli* strains C41(DE3), pBR322 vector with medium copy number, *E. coli* native galU (UGPase) gene, *komagataeibacter* native bacterial cellulose synthase genes bcsA, bcsB, bcsC, and bcsD, accessory genes cmcAx and ccpAx, promoter sequences (T7lac, AraBAD, pCspA), ribosome binding sites, transcriptional terminators, restriction enzymes, T4 DNA ligase, PCR reagents (DNA polymerase, dNTPs, primers), agarose, DNA ladders, gel electrophoresis apparatus, LB broth, LB agar, selective antibiotics (ampicillin), IPTG, L-arabinose, glycerol stocks, glucose, alternative carbon feedstocks, UTP, induction media, phosphate-buffered saline (PBS), distilled water, shaking incubator, static incubator, temperature-controlled incubator, spectrophotometer (BRAND), centrifuge, microcentrifuge tubes, culture flasks, pipettes and sterile pipette tips, biosafety cabinet, autoclave, cellulose pellicle collection tools, analytical balance, and imaging equipment for cellulose characterization.

The full compiled list of parts with their sources is in this sheet, which easily identifies the part with function and its source in tab one of the document.

https://concordiashanghai-my.sharepoint.com/:x/g/personal/claire2027288_concordiashanghai_org/IQDRJRHReSZtOIwVfSVbORqkAeUuAAI94cY19EtaB2a92xc?e=hCPuqW

The purchasing list

2.2 Source and Sequence Information

galU: The *galU* gene was sourced from *Escherichia coli* and corresponds to the GenBank accession number M98830.1. This gene encodes UTP: α -D-glucose-1-phosphate uridylyltransferase. The purification and functional characterization of UGPase, as well as the sequencing of *galU* in *E. coli*, were experimentally validated and published in the *Journal of Bacteriology* in May 1994 (Weissborn et al., 1994). The M98830.1 sequence represents the native *E. coli* coding sequence and has not been codon-optimized for heterologous expression in other organisms.

[pgm](#): The *pgm* gene was derived from *Escherichia coli* K-12 and corresponds to GenBank accession number U08369.1. This gene encodes the enzyme phosphoglucomutase, which catalyzes the interconversion of glucose-1-phosphate and glucose-6-phosphate. The function and genetic characterization of *pgm* in *E. coli* were experimentally confirmed and reported in the *Journal of Bacteriology* in September 1994 (Lu & Kleckner, 1994). The sequence used in this project is the native *E. coli* coding sequence (1,641 bp; nucleotides 121-1761) and has not been codon-optimized for expression in alternative hosts.

[RBS \(BBa_B0030\)](#): BBa_B0030 is a synthetic ribosome binding site designed for use in *E. coli* and other prokaryotic chassis. This part is registered in both the iGEM Registry of Standard Biological Parts and [SynBioHub](#). BBa_B0030 has been widely utilized across multiple iGEM projects, including the 2022 William & Mary iGEM Team's *chassEASE* project, where it was used to drive translation upstream of a GFP coding sequence (*Experiments | William_and_Mary - iGEM 2022*, 2022). Its consistent performance makes it a reliable choice for controlled translation initiation in this system.

[ABT7lacO \(BBa_J435350\)](#): BBa_J435350 is a synthetic T7lac promoter designed for use in *E. coli* and other prokaryotic chassis. It is officially registered in the iGEM Registry of Standard Biological Parts. Inducible under IPTG, the promoter is flanked by GGAG and TACT on its five and third end, respectively.. As a regulatory DNA sequence, the T7lac promoter does not require codon optimization. In this project, BBa_J435350 is incorporated into the plasmid backbone to enable tightly controlled, high-level transcription in DE3-based *E. coli* strains.

[araBAD \(BBa_K808000\)](#): BBa_K808000 is a synthetic arabinose-inducible regulatory unit designed for *E. coli* and other bacterial hosts. It is documented in the iGEM Registry of Standard Biological Parts (Part ID: BBa_K808000 — araC-Pbad arabinose-inducible regulatory promoter/repressor unit) and [SynBioHub](#). The regulatory mechanism of the araBAD promoter was experimentally characterized by Guzman et al. (1995), who demonstrated its tight transcriptional control and strong induction in the presence of L-arabinose. As a regulatory element, the araBAD promoter is not codon-optimized; the *araC* coding sequence remains native to *E. coli*.

[pCspA \(BBa_K4987003\)](#): BBa_K4987003 is a synthetic regulatory part based on the cold-shock-inducible *cspA* promoter from *E. coli*. This promoter is officially registered in the iGEM Registry of Standard Biological Parts as a cold-inducible transcriptional control element. The 2025 BSDFZ China iGEM Team experimentally validated BBa_K4987003, demonstrating strong promoter activation at low temperatures while maintaining minimal basal expression under non-inducing conditions (Results | BSDFZ-CHINA – iGEM 2025, 2025). It is not codon-optimized as pCspA is a non-coding regulatory DNA sequence.

2.3 Design Rationale, Justifications, and Functional Role

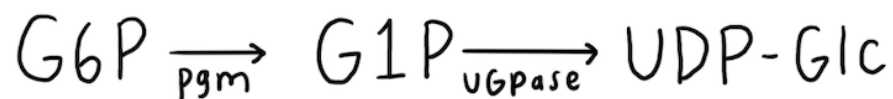


Figure 4. This diagram shows the chemical process of G6P to G1P and to UDP-Glc. Phosphoglucomutase and UGPase are the enzymes involved in this process to catalyze the conversion of glucose.

Genes of Interest

pgm: The *pgm* gene encodes phosphoglucomutase, an enzyme that catalyzes the reversible interconversion of glucose-1-phosphate into glucose-6-phosphate via the enzyme-bound intermediate glucose -1,6-bisphosphate (Lu & Kleckner, 1994). This gene is native and highly conserved in both *E. coli* and *Acetobacter xylinum*. Because *galU* utilizes glucose-1-phosphate as its substrate, co-overexpression of *pgm* is intended to ensure sufficient precursor availability to support increased UDP-Glc synthesis.

galU: The *galU* gene encodes UTP:ox-D-glucose-1-phosphate uridylyltransferase, also known as UDP-glucose pyrophosphorylase (UGPase) (Weissborn et al., 1994). UGPase catalyzes the conversion of glucose-1-phosphate and UTP into UDP-glucose, the direct metabolic precursor for bacterial cellulose synthesis. By overexpressing *galU*, this design aims to increase intracellular UDP-glucose levels and thereby enhance substrate availability for downstream cellulose polymerization by the *bcs* operon.

Promoter Choice

T7lac promoter: The DE3 strains used in this study are based on the T7 system, making T7-derived promoters well-suited for strong, high-level transcription. The use of both the T7 standard promoter and the T7lac promoter enables comparison between constitutive high expression and regulated induction. The T7lac promoter contains a lac operator downstream of the promoter, enabling tight control of expression via IPTG induction (Schuster & Reisch, 2022). This regulation is critical for balancing strong expression with cellular function when overexpressing potentially burdensome metabolic enzymes. This is included in our plasmid vector backbone.

araBAD promoter: araBAD is induced by arabinose, and its activity can be regulated by the amount of the inducer. Since we want to overexpress *pgm* and *galU* in *E. coli*, which can be a metabolic burden for the cell, araBAD promoter has the advantage of being able to repress transcription in the absence of L-arabinose. Moreover, UGPase might be toxic because UGPase can disrupt flux balance when overexpressed, so araBAD promoter is ideal for controlling transcription.

pCspA promoter: pCspA is the promoter of the *cspA* gene, which encodes a cold shock protein in *E. coli*. Upon a temperature downshift, *E. coli* activates a cold shock response that selectively upregulates proteins involved in stabilizing transcription and translation at low temperatures (You, 2023). Leveraging this response, pCspA enables temperature-inducible expression under conditions that are both physiologically favorable for UGPase and BC synthesis and empirically associated with improved recombinant bacterial cellulose production, making it a strategically suited promoter for this system.

Ribosome Binding Site

BBa_B0030 : This RBS is considered as a strong RBS relative to BBa_B0031 and BBa_B0032, (Mahajan et al., 2016) meaning that it can recruit ribosomes more efficiently and drives higher protein expression. Furthermore, Austin-UTexas 2020 found that this RBS exhibits essentially no burden on the cell growth despite high activity, meaning that it was a natural choice. This RBS has shown to work in BL21 (DE3), and is likely to be compatible with C41 (DE3).

Tags and Reporters

GFP: GFP is a reporter protein that is relatively stable, small, and easy to use: It only requires oxygen and an energy source to perform its function (March et al., 2003). By inserting the GFP gene into our cells, we can quantify the amount of the cell, which may imply UDP-Glc production. Additionally, GFP emits a recognizable glow under UV light, making it easy to quickly identify transformation success.

Backbone (terminator):

BBA_B1002: Because a variety of promoters are being tested, a standard, small, and high efficiency terminator was chosen. It is a forward terminator with forward efficiency of 0.98[CH] (Part:BBa B1002 - Parts.igem.org, 2025).

2.3 - Plasmid Design

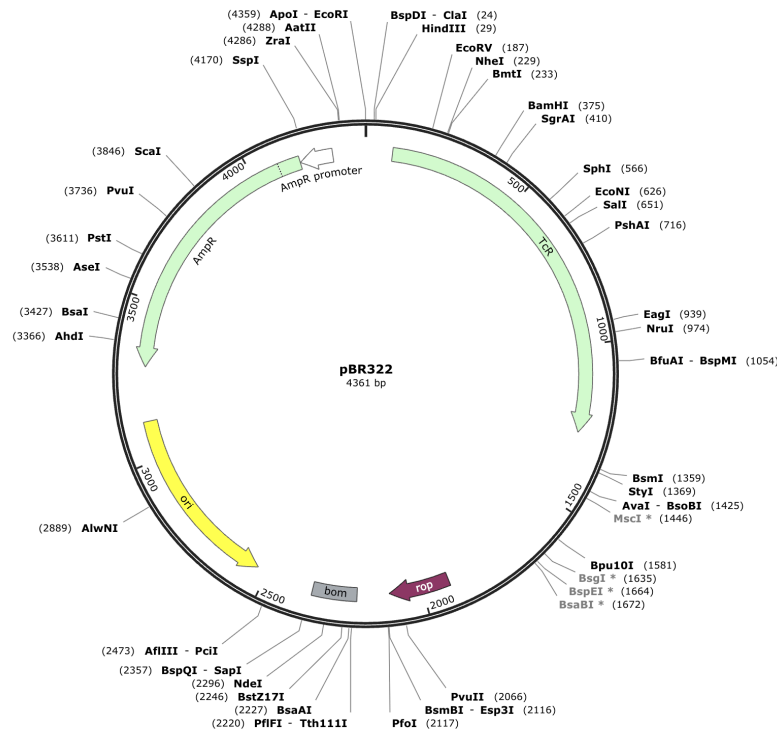


Figure 5. Plasmid backbone for pBR322, sourced by Snapgene.

The pBR322 vector contains ampicillin (AmpR) and tetracycline (TetR) resistance, a high copy origin of replication, and the rop protein that maintains the low copy number. Overall, the vector has a medium copy number of around 20 copies. For easier ligation, we chose a vector that did not contain a promoter. Additionally, the inclusion of the BsaI restriction site is particularly useful in creating complementary sticky ends with the promoter that will be inserted, and this revolved around the T7lac promoter we will insert, which was flanked by BsaI restriction sites. The two resistance genes featured are particularly useful when inserting our sequence into the backbone, as cutting at BsaI and inserting our construct will interfere with the expression of AmpR, which is needed for growth. Because of this, our agar plates will be treated with tetracycline, the alternative resistance marker.

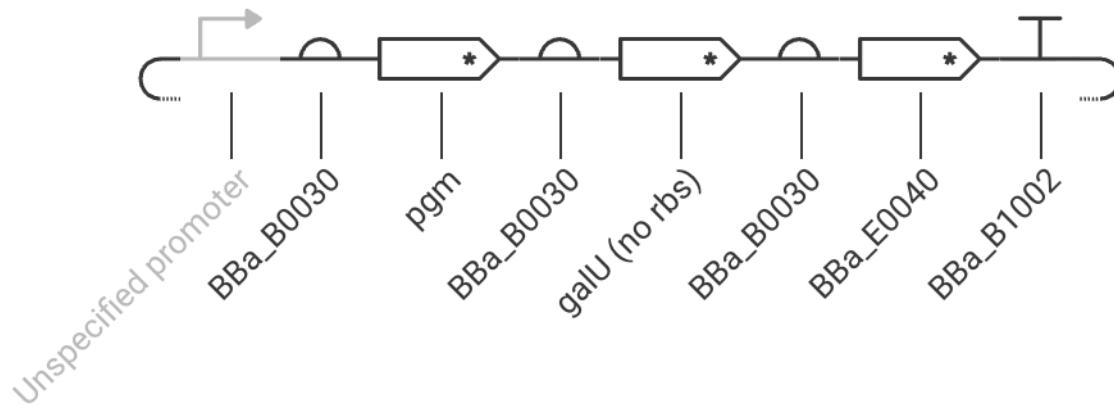


Figure 6. The genetic construct of promoter-GalU-pgm-GFP, which was constructed using Kernel. Asimov, Inc. (2026). Kernel [Version 3.239.3] [Web-based software]. <https://www.asimov.com/kernel> (Accessed January 15, 2026)

3. Phase 1: Homologous overexpression of *galU* and *pgm* in *E. coli*

3.1 Experimental Design

In this phase, we will investigate the effects of different promoters (T7lac, araBAD, and pCspA) on *pgm* and UGPase production in *E. coli*. A GFP coding sequence will be attached downstream of both genes to enable real-time tracking of protein expression using a spectrophotometer.

Phase 1.0: Control Assays

At 37 °C, cells will be transformed with an empty plasmid vector. Following successful transformation, inducer concentrations consistent with phase 1.1 will be added into cell culture. OD600 and GFP measurements will be taken for both wild-type cells and T7lac control cells at 0, 3, and 6 hours. At 6 hours, a *galU* assay will be performed on both cell lines.

Phase 1.1: Inducer Concentration Assay

At 37 °C, an IPTG concentration assay will be conducted on the T7lac/*pgm/galU* cell line using a range of IPTG concentrations (0, 0.05mM, 0.1mM, 0.5 mM, 1mM) In parallel, an arabinose concentration assay will be conducted on the araBAD/*pgm/galU* cell line using a range of arabinose concentrations (0, 0.05mM, 0.5 mM, 1mM, 5mM, 15mM). The pCspA/*pgm/galU* cell line will be incubated without an inducer. OD600 and GFP measurements will be taken for all cell lines and inducer concentrations. *galU* assays will be performed only for the inducer concentration that yields the highest GFP expression in the T7lac/*pgm/galU* and araBAD/*pgm/galU* cell lines.

Phase 1.2: Temperature Assay

Using the optimal inducer concentrations identified in Phase 1.1 for the T7lac/*pgm/galU* and araBAD/*pgm/galU* cell lines, all three cell lines (T7lac/*pgm/galU*, araBAD/*pgm/galU*, and pCspA/*pgm/galU*) will be incubated at 25 °C and 30 °C for a total of 6 hours. OD600 and GFP measurements will be taken at 0, 3, and 6 hours for all cell lines. At 6 hours, a *galU* assay will be performed on the pCspA/*pgm/galU* cell line for both temperature conditions.

3.2 Various Protocols

3.2.1 Plasmid Construction



Ligating promoters:

The pgm/galU sequence (as denoted in figure 6) will be synthesized. In addition to the synthesized strand, a TACT overhang before the first RBS and a 3' GCGA overhang after the terminator. Furthermore, synthesized promoters (pCspA and araBAD) will also have 5' GGAG fusion site and a 3' ATGA site, in accordance with Type IIS Assembly standards.

Ligation of T7/Lac onto device: the AB T7/Lac promoter, provided by the 2025 iGEM distribution kit, has a 3' flanked fusion site TACT. On the pgm/galU sequence, we've designed a ATGA sticky end such that effective ligation can occur.

Ligating device onto vector:

Using 1 uL of BsaI, 1 ug of plasmid, 5 uL of 1X CutSmart buffer, and nuclease-free water of up to 50 uL, incubate the vector for 15 minutes for 37°C. Purify both the vector DNA and galU/pgm DNA using the Monarch® Spin DNA Gel Extraction Kit. To ligate the device onto the plasmid, look below for the ligation procedure.

Ligation Procedure:

DNA ligation procedure for 20 uL reaction uses 2 uL of 10x T4 DNA buffer, 50 ng of vector DNA, 37.5 ng of insert DNA, 1 uL of T4 DNA Ligase, and nuclease-free water. T4 DNA buffer needs to be thawed and resuspended at room temperature. Gently mix the reaction by pipetting up and down, and briefly centrifuge. Incubate at 16°C overnight or room temperature for 10 minutes. Heat inactivate at 65°C for 10 minutes. Run gel electrophoresis to confirm results. Lastly, prior to transformation, chill on ice.

Confirmation of Ligation

- Put the cells in tet. If the cells die then thats good it worked, cells that didn't die -> means that the tet was not successfully disrupted.

References/Tools used:

- Asimov, Inc. (2026). Kernel (Version 3.239.3) [Web-based software]. <https://www.asimov.com/kernel> (Accessed January 15, 2026)
- Vincze, T., Posfai, J., & Roberts, R. J. (2003). NEBcutter: a program to cleave DNA with restriction enzymes. *Nucleic Acids Research**, 31(13), 3688–3691. <https://doi.org/10.1093/nar/gkg526>
- New England Biolabs. (2026). NEBCloner (Version 1.17.0) [Web tool]. <https://nebcloner.neb.com/#/> (Accessed January 15, 2026)

3.2.2 PCR Protocol

Standard PCR protocol will be followed if any parts need amplification. This section will be later expanded on, and primers will also be designed. Our vector with *pgm* and *galU* in it will likely need to be cloned multiple times. Potentially, cell lines such as D5alpha would be an alternative to PCR for vector cloning. A control group will have the plasmid vector and the ligase control.

Amplifying vector

Forward primer: GACTATCGTCGCCGCACTTA

Reverse primer: GCGGTATCATTGCAGCACTG

TM at 60°C.

To perform a standard PCR using Taq DNA polymerase, the reaction mix will be prepared by adding the appropriate reagents into PCR tubes according to the recommended volumes, including PCR buffer, dNTP mix, forward and reverse primers, template DNA, MgCl₂ as needed, and Taq polymerase. After gently mixing and briefly centrifuging the tubes to collect contents at the bottom, the tubes are placed in the PCR machine. Amplification is carried out for 25–30 cycles using the following thermal profile: denature the template at 94 °C for 1 minute to separate the DNA strands, anneal primers at 60 °C for 2 minutes to allow primer binding, and extend at 72 °C for 3 minutes for synthesis of new DNA strands. After cycling, evaluate amplification products by agarose gel electrophoresis with appropriate staining. Mineral oil overlay may be removed if necessary before analysis. The protocol for PCR used a protocol from Sigma Aldrich as a reference (“Standard PCR Protocol,” 2025).

3.2.3 Gel Electrophoresis Protocol

Gel electrophoresis will be performed with negative controls of our gene of interest, an empty pBR322 vector, and buffer solution. The constructed plasmids that will need successful confirmation through the electrophoresis will be plasmids all containing the vector and gene of interest with varying promoters for three plasmids in total.

We will likely need a 1Kbp DNA base pair ladder, as many of the DNA fragments exceed 1Kbp. The protocol for Gel Electrophoresis will follow a simple Agarose base for the gel, which will be made using 1% agarose gel that contains 1 gram of agarose to 100 mL of 0.5 TAE. To make the gel, the agarose is combined with the TAE in a microwaveable flask and is heated up until melted. While waiting for the gel to cool down to around 40~50 C, set up the 9- well comb on the gel box. After cooling, pour the gel into the box, and let it sit at room temperature for around 20-30 minutes. When set, add a buffer solution to the gel box before adding the gel to the box. The DNA samples that will be tested are mixed with 5µl loading dye for every 25µl of DNA, and are mixed well with the tip of a micropipette. From here, a DNA ladder is inserted into the first row, and the wells are loaded with the various DNA samples. When successfully loaded, the gel is run for roughly 40 minutes to separate the DNA fragments.

3.2.4 Transformation Protocol

Prior to transformation, LB agar plates should be prepared. Transformation protocol follows heat shock procedure specified for *E. coli* C41 (DE3) cells, dictated by Lucigen (OverExpress® Chemically Competent Cells, n.d.). Cells are frozen at -86°C, and are thawed completely over wet ice. Harvest cells in their mid-log phase, meaning 0.4-0.6 OD₆₀₀. Prepare ampicillin LB agar plates, with 100µg/mL ampicillin. Add 1 µg of DNA to 50µL of C41 cells on ice, and stir (without pipetting up and down) to mix. For best transformation efficiency, transfer cells into a pre-chilled disposable polypropylene 15-mL culture tube (17 x 100 mm); transforming directly in the container the cells came in will have a lower efficiency. Inoculate on ice for 30 minutes, then heat shock the cells by placing them in a 42°C dry bath for 45 seconds. Note that the protocol asks the cells to be placed in a water bath, but a dry bath would also suffice. Return the ice for 2 minutes. Afterwards, add 950 µL of room-temperature Expression Recovery Medium to the cells in the culture tube, and place the tubes in a shaking incubator at 250 rpm for 1 hour at 37 °C. Then, plate up to 100µL of transformed cells on LB agar plates with ampicillin, and incubate the plates overnight at 37°C. Transformed clones can then be grown in LB.

3.3 GFP Expression Assay Protocol/OD600

The stock solutions are prepared with a constant volume of 5mL of cell suspension, and 100 µL of 0.2mM inducer. From here, the 96 well plates are pipetted with 0.10 mL (100 µL) of solution that was prepped beforehand. Afterwards, the fluorescence and the OD₆₀₀ of the samples will be measured with spectroscopy (GFP emission wavelength is 518 nanometers; OD₆₀₀ uses 600 nm absorbance)(Part:BBa K1911005 - Parts.igem.org, n.d.). Blanks will be created using the GFP LB solution. Once the cells are filled, we will measure the fluorescence of the samples using a spectrophotometer and create an average by rows. The OD₆₀₀ and fluorescence raw data will be collected by a plate reader software before analysis.

3.4 UGPase (GalU)/pgm Assay

Exponentially growing cells were harvested after 6 hours of growth, resuspended in 1/50 of the volume of the original culture. For this assay, we will use a UGPase Assay kit by Xuan Yu Biological Technology Company (上海烜雅生物科技有限公司). The kit is described as a UDPG pyrophosphorylase (UPG) test kit for 48 or 96 samples (depending on the desired amount) and can be used on both tissues and bacteria.

The UV spectrophotometer/enzyme analyzer should be preheated first for 30 minutes. Then, adjust the wavelength to 340nm and zero the instrument with distilled water. Take a micro quartz cuvette/96-well plate, and add 100 µL of reagent one, 20 µL of reagent two, 20 µL of reagent three, 20 µL of reagent four, 20 µL of reagent five, and 20 µL of crude enzyme solution in sequence. Mix well and record the absorbance values A₁ at 30 seconds and A₂ at 330 seconds at 340nm. $\Delta A = A_2 - A_1$

Per the kit's protocols, we will measure UGPase activity by measuring the consumption of 1 nmol NADP per minute by every 10⁴ cells using the calculation formula:

$$\text{UGP (nmol/min / 10}^4 \text{ cell)} = \Delta A \div (\epsilon \times d) \times V_{\text{total}} \div (V_{\text{sample}} \times \text{cell number} \div V_{\text{sample-total}}) \div T = 321.54 \times \Delta A \div \text{cell number}$$

Where V_{total} is the total volume of the reaction system (0.2mL); ϵ is the molar extinction coefficient of NADPH (6.22×10³ L / mol / cm); d is the light path of the cuvette (0.5 cm); V_{sample} is the volume of the sample added (0.02 mL); V_{sample-total} is the volume of the extract added (1 mL); T is the reaction time (5 min).

This will model both UGPase/pgm activity and UDP-Glc synthesis as these reactions are indirectly linked to NADP⁺/NADPH metabolism. High NADP⁺ or low NADPH levels means an increased flux toward UDP-Glc synthesis, as glc-6 dehydrogenase utilizes NADP⁺ to convert G6P into UDP-Glc.

3.5 Data Analysis

For the GFP Expression Assay, all of the raw data values should first be blank corrected with a solution with LB and antibiotic resistance for both the OD₆₀₀ and GFP fluorescence values. Using a standard curve with known fluorescence values using fluorescein, the spectroscopic readings of fluorescence of our cells will be converted into the molecular equivalent of fluorescence (MEFL). This will indirectly illustrate the protein concentrations for our cells. Similarly, a particle standard curve for the OD₆₀₀ to illustrate cell concentration will be a benchmark for cell growth. This will be used to discuss how expression of *pgm* and *galU* genes impacts cell metabolism and growth in varying experimental conditions, allowing us to identify the best promoter and experimental conditions for best protein activity with the least burden on the cells. This will also be used to discuss which promoter and experimental conditions will be best suited for coexpression with BC synthase.

Following the experiments, there will be graphs with the line of best fit illustrating the fluorescence over time in different experimental conditions will be created with the raw data collected by the spectrophotometer. To analyze the significance between groups, R² values will be added to each line of fit to illustrate the proportion of variation predicted by the model, where a value greater than 0.05 will indicate a reasonable difference between the fitted model and the raw data. Additionally, we will use an Analysis of Covariance (ANCOVA) to illustrate a statistically significant difference between the slope of the lines of best fit, which would show how the experimental factors, or the covariants, impact expression levels (Analysis of Covariance - an Overview | ScienceDirect Topics, n.d.). However, unlike ACOVA, this is difficult to do by hand, and it is likely that we will have to use R to model this.

For the UGPase/pgm assay, all samples will be analyzed after six hours of incubation after adding the respective inducer. After adding the reaction mixtures per assay kit, the formation of NADPH is recorded at 37°C at 340 nm until a maximum optimal density is achieved. Assuming that the relationship between pgm and UGPase activity in expressing UDP-Glc directly correlates with the redox-reaction between NADP⁺ and NADPH, we can calculate the concentration of NADPH present using Beer Lambert's law. The molar absorbance constant for NADPH is known to be 6.22 l/mmol/cm (Veskoukis et al., 2017), and the spectrophotometer that will be used will be set to a path length of 1 cm. Using this formula, we can calculate the concentration of NADPH, and thus, UDP-Glc to show the varying levels of GalU and UGPase expression under different promoters in optimal conditions.

A bar graph comparing the experimental groups (Control, T7lac, araBAD, pCspA) and calculated concentrations of NADPH/UDP-Glc expression will be used to identify the optimum time, temperature, and inducer concentrations for the synthesis of UDP-Glc. Error bars showing 95% confidence intervals will be included to represent the precision of the estimated means. Additionally, we will perform a statistical analysis using ANOVA for the bar graphs, where the null hypothesis states that there is no statistical difference between the experimental groups. This will illustrate if the means of the experimental groups are statistically different, thus illustrating the distinction between expression levels in the negative control and the groups due to the optimized conditions. The specific calculations will be done following procedure by the University of Notre Dame (Sedgwick, 2012) (<https://academicweb.nd.edu/~rwilliam/stats1/x52.pdf>).

When comparing the experimental groups to the negative control, higher levels of UDP-Glc production in the recombinant groups will constitute success, as it illustrates the impact of the selected parts and optimized conditions on UGPase/pgm activity, and also for potential future BC experiments.

4. Phase 2: Theoretical Heterogenous expression of BcsABCD in E coli

4.1 Experimental Hypotheses

In phase two, we will address increasing heterogeneous expression of BcsABCD in *E coli* by increasing UDP-glucose levels through the coexpression of these genes with recombinant UGPase in its optimum expression conditions, as found in phase 1. The plasmid containing BcsABCD will use the pACYC184 vector, as it has genes for Chloramphenicol resistance, and a p15A origin of replication, which are different from the pBR322 backbone, allowing for the expression of both plasmids in a single C41 (DE3) cell. This backbone is also compatible with strains that are a BL21 derivative, making this our top choice for the BcsABCD vector.

Based on the promoter from phase 1 that shows the most UGPase expression, we will use the same promoter to allow for a standard inducer and expression conditions. The negative control will contain the BcsABCD plasmid without pBR322-GalU-pgm-GFP, while the experimental groups will have BcsABCD with pBR322-GalU-pgm-GFP, and BcsABCD with the addition of the accessory genes cmcax and CcpAx in pBR322-GalU-pgm-GFP. This will study the impact of the addition of pBR322-GalU-pgm-GFP, and the accessory genes.

4.2 Data Collection and Analysis

After the plasmids are constructed, gel electrophoresis will be performed to verify the successful assembly of the plasmid as described in section 3.2 with the constructed plasmids, an empty backbone, the accessory genes, and the BcsABCD construct.

After successful transformation of the plasmids into the cells, as outlined in 3.2.4, the cells will be induced with the optimum concentration and optimum temperature found in phase 1. Following this, the dry weight of cellulose that is produced by the experimental groups will be measured.

To measure the dry weight of the cellulose, the cellulose pellets produced on the surface of the growth stock will be washed with distilled water before being filtered. From here, a 1% NaOH solution will be used to kill off the remaining cells. The purified cellulose is rinsed again before being placed in the oven for 1-2 hours at 105 degrees celcius.

(Almihyawi et al., 2024). After the samples are dried, their respective weights will be represented by a bar graph with 95% Confidence interval error bars. Moreover, a statistical analysis using ANOVA will be performed to illustrate that the differences between the experimental groups are significant, and similar to section 3.5 in phase 1. The experimental group that shows the greatest dry weight of BC will be compared to the control group to illustrate that successful overexpression of UGPase and UDP-Glc will increase BC synthesis.

References

- Abeer, M. M., Mohd Amin, M. C. I., & Martin, C. (2014). A review of bacterial cellulose-based drug delivery systems: their biochemistry, current approaches and future prospects. *Journal of Pharmacy and Pharmacology*, 66(8), 1047–1061. <https://doi.org/10.1111/jphp.12234>
- Addgene. (2016, September 14). Pouring LB agar plates. Addgene.org. <https://www.addgene.org/protocols/pouring-lb-agar-plates/>
- Abidi, W., Lucía Torres-Sánchez, Axel Siroy, & Krasteva, P. V. (2021). Weaving of bacterial cellulose by the Bcs secretion systems. *Fems Microbiology Reviews*, 46(2). <https://doi.org/10.1093/femsre/fuab051>
- Acheson, J. F., Ho, R., Goularte, N. F., Cegelski, L., & Zimmer, J. (2021). Molecular organization of the E. coli cellulose synthase macrocomplex. *Nature Structural & Molecular Biology*, 28(3), 310–318. <https://doi.org/10.1038/s41594-021-00569-7>
- Almihyawi, R. A. H., Musazade, E., Alhussany, N., Zhang, S., & Chen, H. (2024). Production and characterization of bacterial cellulose by Rhizobium sp. isolated from bean root. *Scientific Reports*, 14(1), 10848. <https://doi.org/10.1038/s41598-024-61619-w>
- Analysis of Covariance - an overview | ScienceDirect Topics. (n.d.). Wwww.sciencedirect.com. <https://www.sciencedirect.com/topics/medicine-and-dentistry/analysis-of-covariance>
- Aswini, K., Gopal, N. O., & Uthandi, S. (2020). Optimized culture conditions for bacterial cellulose production by *Acetobacter senegalensis* MA1. *BMC Biotechnology*, 20(1). <https://doi.org/10.1186/s12896-020-00639-6b>
- Bedore, A. M., & Waters, C. M. (2024). Plasmid-free cheater cells commonly evolve during laboratory growth. *Applied and environmental microbiology*, 90(4), e0231123. <https://doi.org/10.1128/aem.02311-23>
- Buldum, G., Bismarck, A., & Mantalaris, A. (2018). Recombinant biosynthesis of bacterial cellulose in genetically modified *Escherichia coli*. *Bioprocess and Biosystems Engineering*, 41(2), 265–279. <https://doi.org/10.1007/s00449-017-1864-1>
- Buldum, G., & Mantalaris, A. (2021). Systematic Understanding of Recent Developments in Bacterial Cellulose Biosynthesis at Genetic, Bioprocess and Product Levels. *International Journal of Molecular Sciences*, 22(13), 7192. <https://doi.org/10.3390/ijms22137192>
- Dumon-Seignovert, L., Cariot, G., & Vuillard, L. (2004). The toxicity of recombinant proteins in *Escherichia coli*: a comparison of overexpression in BL21(DE3), C41(DE3), and C43(DE3). *Protein Expression and Purification*, 37(1), 203–206. <https://doi.org/10.1016/j.pep.2004.04.025>
- Experiments | William_and_Mary - iGEM 2022. (2022). Igem.wiki. <https://2022.igem.wiki/william-and-mary/experiments>
- Fang, J., Kawano, S., Tajima, K., & Kondo, T. (2015). In Vivo Curdlan/Cellulose Bionanocomposite Synthesis by Genetically Modified *Gluconacetobacter xylinus*. *Biomacromolecules*, 16(10), 3154–3160. <https://doi.org/10.1021/acs.biomac.5b01075>
- Florea, M., Hagemann, H., Santosa, G., Abbott, J., Micklem, C. N., Spencer-Milnes, X., de Arroyo Garcia, L., Paschou, D., Lazenbatt, C., Kong, D., Chughtai, H., Jensen, K., Freemont, P. S., Kitney, R., Reeve, B., & Ellis, T. (2016). Engineering control of bacterial cellulose production using a genetic toolkit and a new cellulose-producing strain. *Proceedings of the National Academy of Sciences*, 113(24), E3431–E3440. <https://doi.org/10.1073/pnas.1522985113>

- Fillat, A., Martínez, J., Valls, C., Cusola, O., Roncero, M. B., Vidal, T., Valenzuela, S. V., Diaz, P., & Pastor, F. I. J. (2018). Bacterial cellulose for increasing barrier properties of paper products. *Cellulose*, 25(10), 6093–6105. <https://doi.org/10.1007/s10570-018-1967-0>
- Guzman, L. M., Belin, D., Carson, M. J., & Beckwith, J. (1995). Tight regulation, modulation, and high-level expression by vectors containing the arabinose PBAD promoter. *Journal of Bacteriology*, 177(14), 4121–4130. <https://doi.org/10.1128/jb.177.14.4121-4130.1995>
- Huang, L., Zhou, Y., Feng, Y., Jia, S., Wang, S., & Zhong, C. (2025). Tailoring Bacterial Cellulose through the CRISPR/Cas9-Mediated Gene Editing Tool in *Komagataeibacter xylinus*. *ACS synthetic biology*, 14(4), 1161–1172. <https://doi.org/10.1021/acssynbio.4c00785>
- Imai, T., Sun, S., Horikawa, Y., Wada, M., & Sugiyama, J. (2014). Functional Reconstitution of Cellulose Synthase in *Escherichia coli*. *Biomacromolecules*, 15(11), 4206–4213. <https://doi.org/10.1021/bm501217g>
- Jacek, P., Kubiak, K., Ryngajło, M., Rytczak, P., Paluch, P., & Bielecki, S. (2019). Modification of bacterial nanocellulose properties through mutation of motility related genes in *Komagataeibacter hansenii* ATCC 53582. *New Biotechnology*, 52, 60–68. <https://doi.org/10.1016/j.nbt.2019.05.004>
- Jong In Park, Takeshi Ishimizu, Keita Suwabe, Sudo, K., Hiromi Masuko, Hirokazu Hakozaiki, Ill Sup Nou, Suzuki, G., & Watanabe, M. (2010). UDP-Glucose Pyrophosphorylase is Rate Limiting in Vegetative and Reproductive Phases in *Arabidopsis thaliana*. *Plant and Cell Physiology*, 51(6), 981–996. <https://doi.org/10.1093/pcp/pcq057>
- Kuo, C.-H., Teng, H.-Y., & Lee, C.-K. (2015). Knock-out of glucose dehydrogenase gene in *Gluconacetobacter xylinus* for bacterial cellulose production enhancement. *Biotechnology and Bioprocess Engineering*, 20(1), 18–25. <https://doi.org/10.1007/s12257-014-0316-x>
- Li, G., Wang, L., Deng, Y., & Wei, Q. (2021). Research progress of the biosynthetic strains and pathways of bacterial cellulose. *Journal of Industrial Microbiology and Biotechnology*. <https://doi.org/10.1093/jimb/kuab071>
- Liu, M., Li, S., Xie, Y., Jia, S., Hou, Y., Zou, Y., & Zhong, C. (2018). Enhanced bacterial cellulose production by *Gluconacetobacter xylinus* via expression of *Vitreoscilla* hemoglobin and oxygen tension regulation. *Applied microbiology and biotechnology*, 102(3), 1155–1165. <https://doi.org/10.1007/s00253-017-8680-z>
- Lu, M., & Kleckner, N. (1994). Molecular cloning and characterization of the pgm gene encoding phosphoglucomutase of *Escherichia coli*. *Journal of Bacteriology*, 176(18), 5847–5851. <https://doi.org/10.1128/jb.176.18.5847-5851.1994>
- Naomi, R., Bt Hj Idrus, R., & Fauzi, M. B. (2020). Plant- vs. Bacterial-Derived Cellulose for Wound Healing: A Review. *International Journal of Environmental Research and Public Health*, 17(18), 6803. <https://doi.org/10.3390/ijerph17186803>
- Mahajan, V. S., Marinescu, V. D., Chow, B., Wissner-Gross, A. D., & Carr, P. (2016). *Part:BBa B0030 - parts.igem.org*. Igem.org. https://parts.igem.org/wiki/index.php/Part:BBa_B0030
- Malcı, K., Li, I. S., Kisseroudis, N., & Ellis, T. (2024). Modulating Microbial Materials - Engineering Bacterial Cellulose with Synthetic Biology. *ACS Synthetic Biology*. <https://doi.org/10.1021/acssynbio.4c00615>
- March, J. C., Rao, G., & Bentley, W. E. (2003). Biotechnological applications of green fluorescent protein. *Applied Microbiology and Biotechnology*, 62(4), 303–315. <https://doi.org/10.1007/s00253-003-1339-y>

- Mazhar Ul-Islam, Khan, S., Muhammad Wajid Ullah, & Do Joong Park. (2019). Comparative Study of Plant and Bacterial Cellulose Pellicles Regenerated from Dissolved States. *International Journal of Biological Macromolecules*, 137, 247–252. <https://doi.org/10.1016/j.ijbiomac.2019.06.232>
- Omadjela, O., Narahari, A., Strumillo, J., Mérida, H., Mazur, O., Bulone, V., & Zimmer, J. (2013). BcsA and BcsB form the catalytically active core of bacterial cellulose synthase sufficient for in vitro cellulose synthesis. *Proceedings of the National Academy of Sciences*, 110(44), 17856–17861. <https://doi.org/10.1073/pnas.1314063110>
- OverExpress® Chemically Competent cells. (n.d.). Retrieved January 15, 2026, from https://biosearchtech.a.bigcontent.io/v1/static/manual_COMCEL-004_OverExpress-Chemicallycompetent-Cells
- Pang, M., Huang, Y., Meng, F., Zhuang, Y., Liu, H., Du, M., Ma, Q., Wang, Q., Chen, Z., Chen, L., Cai, T., & Cai, Y. (2020). Application of bacterial cellulose in skin and bone tissue engineering. *European Polymer Journal*, 122, 109365. <https://doi.org/10.1016/j.eurpolymj.2019.109365>
- Part:BBa B1002 - parts.igem.org. (2025). Igem.org. https://parts.igem.org/wiki/index.php?title=Part:BBa_B1002
- Part:BBa K1911005 - parts.igem.org. (n.d.). Parts.igem.org. https://parts.igem.org/Part:BBa_K1911005
- Popa, L., Ghica, M. V., Tudoroiu, E.-E., Ionescu, D.-G., & Dinu-Pîrvu, C.-E. (2022). Bacterial Cellulose—A Remarkable Polymer as a Source for Biomaterials Tailoring. *Materials*, 15(3), 1054. <https://doi.org/10.3390/ma15031054>
- Pourrameza, G. Z., Roayaei, A. M., & Qezelbash, Q. R. (2009). Optimization of Culture Conditions for Bacterial Cellulose Production by *Acetobacter* sp. 4B-2. *Biotechnology(Faisalabad)*, 8(1), 150–154. <https://doi.org/10.3923/biotech.2009.150.154>
- Raghavendran, V., Asare, E., & Roy, I. (2020). Bacterial cellulose: Biosynthesis, production, and applications. *Advances in Microbial Physiology*, 89–138. <https://doi.org/10.1016/bs.amphs.2020.07.002>
- Results | BSDFZ-CHINA - iGEM 2025. (2025). Igem.wiki. <https://2025.igem.wiki/bsdfz-china/results>
- Schuster, L. A., & Reisch, C. R. (2022). Plasmids for Controlled and Tunable High-Level Expression in *E. coli*. *Applied and Environmental Microbiology*, 88(22), e0093922. <https://doi.org/10.1128/aem.00939-22>
- Sedgwick, P. (2012). One way analysis of variance. *BMJ*, 344(apr04 1), e2427–e2427. <https://doi.org/10.1136/bmj.e2427>
- Shigematsu, T., Takamine, K., Kitazato, M., Morita, T., Naritomi, T., Morimura, S., & Kida, K. (2005). Cellulose production from glucose using a glucose dehydrogenase gene (gdh)-deficient mutant of *Gluconacetobacter xylinus* and its use for bioconversion of sweet potato pulp. *Journal of bioscience and bioengineering*, 99(4), 415–422. <https://doi.org/10.1263/jbb.99.415>
- Sid, S., Mor, R. S., Kishore, A., & Sharanagat, V. S. (2021). Bio-sourced polymers as alternatives to conventional food packaging materials: A review. *Trends in Food Science & Technology*, 115, 87–104. <https://doi.org/10.1016/j.tifs.2021.06.026>
- Sigma - Aldrich. (n.d.). *pETDuet-1 DNA* - Novagen. www.sigmaaldrich.com. <https://www.sigmaaldrich.com/US/en/product/mm/71146?srsId=AfmBOorjrzL-NKerDdLKL0iPGKdcEIUWZJBuxqiVPINTMWB4GVFA-m1>

- Singhania, R. R., Patel, A. K., Tsai, M.-L., Chen, C.-W., & Di Dong, C. (2021). Genetic modification for enhancing bacterial cellulose production and its applications. *Bioengineered*, 12(1), 6793–6807.
<https://doi.org/10.1080/21655979.2021.1968989>
- Standard PCR Protocol. (2025). Merck, 1(1).
https://www.sigmaaldrich.com/US/en/technical-documents/protocol/genomics/pcr/standard-pcr?srsId=AfmBOoq0cLgXHV_ar0EpzzgTLgm8dsxvUIyLVaHjENoxmbxCw4TAnJPQ
- Swingler, S., Gupta, A., Gibson, H., Kowalczyk, M., Heaselgrave, W., & Radecka, I. (2021). Recent Advances and Applications of Bacterial Cellulose in Biomedicine. *Polymers*, 13(3), 412.
<https://doi.org/10.3390/polym13030412>
- Teh, M. Y., Ooi, K. H., Danny Teo, S. X., Bin Mansoor, M. E., Shaun Lim, W. Z., & Tan, M. H. (2019). An Expanded Synthetic Biology Toolkit for Gene Expression Control in Acetobacteraceae. *ACS synthetic biology*, 8(4), 708–723. <https://doi.org/10.1021/acssynbio.8b00168>
- Tepap, C. Z., Anissi, J., & Bounou, S. (2023). Recent strategies to achieve high production yield of recombinant protein: A review. *Journal of Cellular Biotechnology*, 1–13. <https://doi.org/10.3233/jcb-220084>
- Trigona, C., Graziani, S., Di Pasquale, G., Pollicino, A., Nisi, R., & Licciulli, A. (2019). Green Energy Harvester from Vibrations Based on Bacterial Cellulose. *Sensors*, 20(1), 136. <https://doi.org/10.3390/s20010136>
- Tronser, T., Laromaine, A., Roig, A., & Levkin, P. A. (2018). Bacterial Cellulose Promotes Long-Term Stemness of mESC. *ACS Applied Materials & Interfaces*, 10(19), 16260–16269. <https://doi.org/10.1021/acsami.8b01992>
- Veskoukis, A. S., Margaritelis, N. V., Kyparos, A., Paschalis, V., & Nikolaidis, M. G. (2017). Spectrophotometric assays for measuring redox biomarkers in blood and tissues: the NADPH network. *Redox Report*, 23(1), 47–56.
<https://doi.org/10.1080/13510002.2017.1392695>
- Weissborn, A. C., Liu, Q., Rumley, M. K., & Kennedy, E. P. (1994). UTP: alpha-D-glucose-1-phosphate uridylyltransferase of Escherichia coli: isolation and DNA sequence of the galU gene and purification of the enzyme. *Journal of Bacteriology*, 176(9), 2611–2618. <https://doi.org/10.1128/jb.176.9.2611-2618.1994>
- Wiem Abidi, Lucía Torres-Sánchez, Axel Siroy, Petya Violinova Krasteva, Weaving of bacterial cellulose by the Bcs secretion systems, *FEMS Microbiology Reviews*, Volume 46, Issue 2, March 2022, fuab051,
<https://doi.org/10.1093/femsre/fuab051>
- Yadav, V., Paniliatis, B. J., Shi, H., Lee, K., Cebe, P., & Kaplan, D. L. (2010). Novel in vivo-degradable cellulose-chitin copolymer from metabolically engineered Gluconacetobacter xylinus. *Applied and environmental microbiology*, 76(18), 6257–6265. <https://doi.org/10.1128/AEM.00698-10>
- You, Y. (2023). *Part:BBa_K4907008*. Igem.org. https://parts.igem.org/Part:BBa_K4907008
- Zhang, H., Ye, C., Xu, N., Chen, C., Chen, X., Yuan, F., Xu, Y., Yang, J., & Sun, D. (2017). Reconstruction of a Genome-scale Metabolic Network of Komagataeibacter nataicola RZS01 for Cellulose Production. *Scientific Reports*, 7(1). <https://doi.org/10.1038/s41598-017-06918-1>
- Zywicka, A., Dunisławska, A., & Fijalkowski, K. (2025). Increased Expression of Bacterial Cellulose Synthase Genes in *Komagataeibacter Xylinus* Exposed to a Rotating Magnetic Field. *Biomacromolecules*, 26(10), 6542–6551.
<https://doi.org/10.1021/acs.biomac.5c00653>